

ALLIANCE KCSE 2025 PREDICTION



POSSIBLE KCSE EXAM



BIOLOGY

231/3

PAPER 3

(PRACTICAL)

TIME: 1¾ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

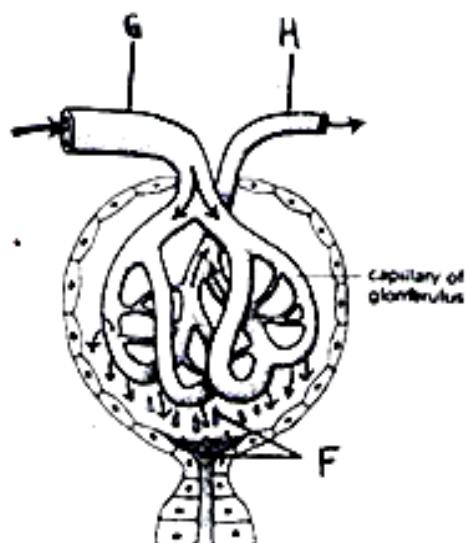
INSTRUCTIONS TO CANDIDATES

- 1. Write your name, class and admission number in the spaces provided above.*
- 2. Sign and write the date of the examination in the spaces provided.*
- 3. Answer all questions in the spaces provided.*
- 4. You are required to spend the first 15 minutes of the 1 ¾ hours allowed for this paper reading the whole paper carefully before commencing your work.*
- 5. Additional pages must not be inserted.*
- 6. Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing*

FOR EXAMINERS USE ONLY

Section	Maximum Marks	Candidate Score
1	13	
2	16	
3	11	
TOTAL SCORE	40	

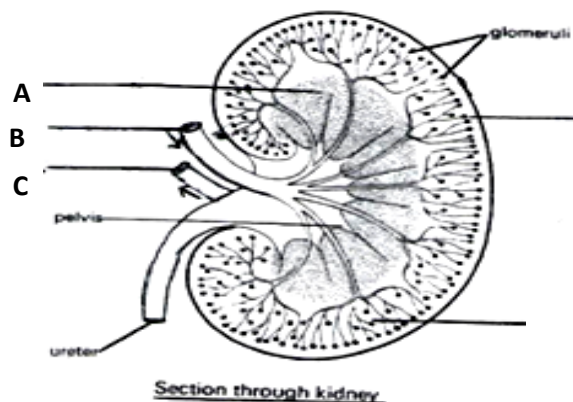
1. The diagram below shows structures which occur in a mammalian kidney. Study the figure carefully and answer the questions that follow.



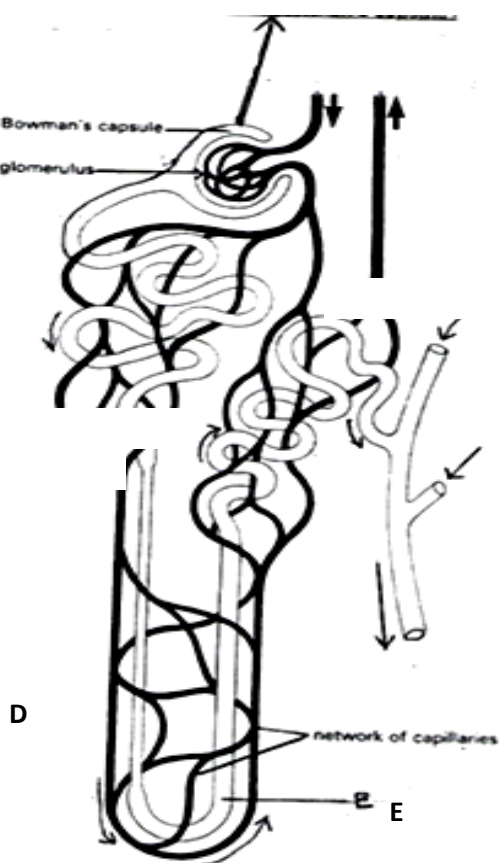
Detail of bowman's capsule

Capillary of glomerulus

Glomerulus



Section through kidney



Network of capsule

Simplified diagram of a kidney tubule

i) Name the parts marked (2 marks)

A.....

D.....

ii) Identify the blood vessels labeled: (4 marks)

B.....

C.....

G.....

H.....

b) i) Name the filtrate labeled F (1 mark)

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ii) Briefly explain how the filtrate named in (b) (i) above is formed (3 marks)

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c) Explain the difference in structure E in a desert and a fresh water animal. (2 marks)

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d) Name the hormone which regulates reabsorption of sodium ions in the kidney tubules. (1 mark)

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2. The diagrams below represent leaves of certain plants



(a) Use the above specimen to complete the dichotomous key below (2marks)

1. (a) Leaf simple..... go to 2
- (b) Leaf compound..... go to 4
2. (a) leaf with parallel veins..... wandering jew
- (b) Leaf with net veins..... go to 3

3. (a) leaf with smooth margin..... Devils horse whip
 (b) Leaf with Tick berry
4. (a) Leaf trifoliate..... go to 5
 (b) Leaf with more than three leaflets..... go to 6
5. (a) Leaf with sharp tips..... Bean
 (b) Leaf with rounded tips..... Oxalis
6. (a) Leaf pinnate..... .Cassia
 (b) Leaf Accacia

(b) Use the dichotomous key above to fill the table below

(14marks)

Specimen	Steps	Identity

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3. You are provided with a specimen labeled K and solution labeled P and Q. Cut the specimen into two halves.

(a) (i) Name the type of reproduction exhibited by specimen K **(1mark)**

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(ii) Give a reason for your answer in a(i) above **(1mark)**

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.....

Using specimen K, remove some of the inner leaves. Cut the leaves along their lengths into nine strips equal length. Each strip should be about 2mm wide. Place three strips into the solution labeled P, place another three strips into the solution labeled Q and leave the last three strips in a petri dish labeled R. Allow the experiment setup to stand for 10 minutes.

(b) Use your fingers to feel the texture of the strips. Record your observations **(2marks)**

i) Strips in solution P

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ii) Strips in solution Q

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(c) Account for the texture of the strips in the solution Q **(4marks)**

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(d) Suggest the concentration of solution P in relation to the cell sap in the strips of the specimen. Give a reason for your answer **(1mark)**

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(e) State the aim of the setup R **(1mark)**

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